

1 Information and Communication Technology

15 April 2023

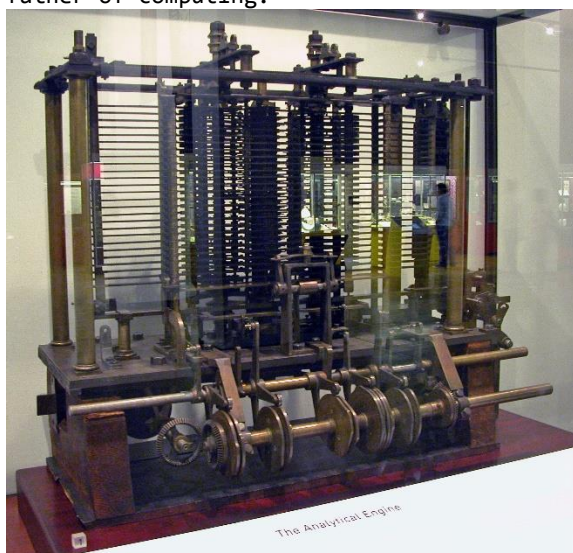
Notes

1. The numbers, words, images and symbols which do not bear a meaning, when standing alone are called data.
2. Data can be classified as
 - a. Quantitative Data - Where data can be represented as numerical values are known as quantitative data. With this data we can do arithmetic calculations like addition, subtraction, sorting and etc.
 - b. Qualitative Data - This can be maps, documents, etc but cannot be represented with numerical values.
3. Data can be
 - a. Collected and saved.
 - b. Can be taken to a different place.
 - c. Can be processed.
4. Essential components of system.
Data -> Process -> Information
5. Data Processing can
 - a. Be processed manually - all the process done manually
 - b. Mechanical Data Processing - Data processing happens with Mechanical Data Processing. Though once popular due to the advancement of Electronic Data processing lost it's popularity.
 - c. Electronic Data Processing - This is the technique often used today's data processing using computers or electronic equipment's. Compared to the manual data processing speed of processing the data is much quicker.
6. Quality of Data
7. Application of Information Technology
 - a. e-government
 - i. **G2C** - Government to Citizen
 1. Students can view their results and order the certificates via online.
 2. Birth certificates can be obtained from the DS offices irrespective of the place of birth.
 3. National Water supply and drainage board water bills can be settled via online.
 4. Vehicle revenue license can be paid online.
 - ii. **G2G** - Government to Government
 1. Diplomatic and Visa Information
 2. Government Law
 - iii. **G2B** - Government to Business
 1. Payment Services
 2. Business Registration
 - iv. **G2E** - Government to Employees
 1. Gazettes
 2. Circulars
 - b. Information Technology in Education
 - i. Class timetables, teacher timetables, room allocation can be obtained through computer software. Teacher substitutes (in case of absent) can be managed through computers.
 - ii. Can study from anyplace from anywhere at any time.
 1. Websites like <https://e-thaksalawa.moe.gov.lk/> provide study materials and past papers in one place.
 2. Student can makes of Microsoft forms/Google Forms to do the automate the MCQ type questions like <https://ictmcq.home.blog/> especially for ICT subject.
 - iii. Serves as a teaching aid for the teacher.
 1. Teachers and students can makes of the animation to understand the concepts that are hard to understand.
<https://phet.colorado.edu/en/simulations/translated/en>

iv. LMS (Learning Management System)

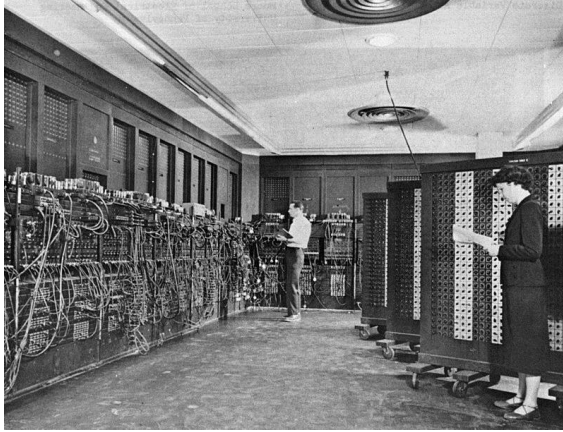
For a student	For school management
The ability to use the learning units at anytime and anywhere at school, at home or at any other convenient place.	The ability to add quality learning materials
The ability to upload the assignments completed at home.	Supervision of activities and publishers of results.
The ability to submit queries, getting replies and submitting comments through Forums	The ability to maintain updated information of the students, teachers and school
The parents can monitor the progress of their children from home.	School can keep track of the progress of the students.

5. Higher Education for every one
 - a. Online assignment and quizzes.
3. In Telemedicine
 - a. Medically qualified person needs to assist the patient while a link established with a Medical expert offsite. This is useful if is a remote area or traveling in a ship.
4. In Agriculture
 - a. Automated drip irrigation.
5. In Livestock development
 - a. Automated Incubator
6. In Fishery
 - a. Mark the best places with the help of GPS comes back to the same place.
7. In manufacturing Industry and Business
 - a. CAD - Computer Aided Design
 - b. CAM - Computer Aided Manufacturing
 - c. Video conferencing - Skype, Zoom, Teams. Viber for group calls/video calls.
8. Demerits/disadvantages of Information Technology
 - a. Eye strain - by using computers or mobile devices continuously eye stain will occur.
 - b. Addicted to Computer Games.- Waste time and increases obesity.
9. Evolution of Computers
 - a. Charles Babbage invented the analytic engine using punch card system concepts. The mechanical machine is based on Input, Process and output concepts. Since this concept helped the development of the computers he is considered as a father of computing.



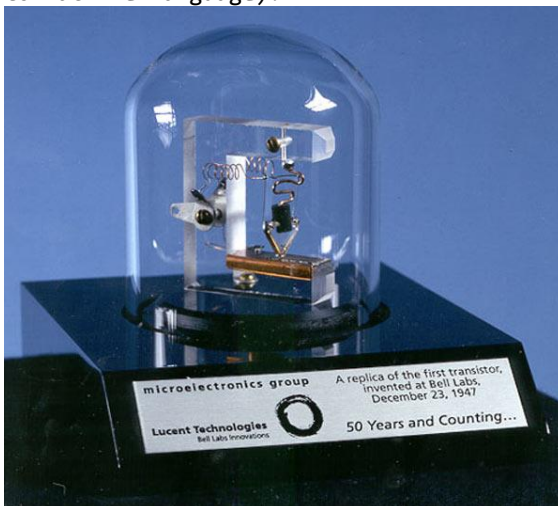
2. Ada Augusta Lovelace is considered as the first programmer.
10. First generation computers were built
 - a. Used Vacuum tubes.
 - b. Machine Languages were used.
 - c. Huge Size .
 - d. Requires more electricity
 - e. Generated more heat.
 - f. Expensive

- g. Punch cards were used for input, output and storage.



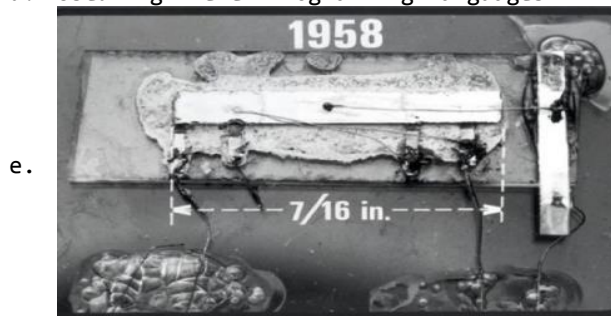
11. Second generation Computers

- a. Were built using Transistors
- b. Used Assembly Languages (requires an assembler to convert the assembly language to machine language).



12. Third Generation Computers

- a. Were based Integrated Circuits.
- b. Keyboard and mouse used for input.
- c. Operating System was born
- d. Used High Level Programming Languages



13. Fourth Generation Computers

- a. Were based on Microprocessors
- b. Based on LSI (Large Scale Integration - means between 500-300,000 components per chip) and VLSI (Very Large Scale Integration - means more than 300, 000 components per chip).

- c. GUI (Graphical User Interface) were introduced to the OS (Operating System).
14. Fifth Generation Computers
- Were based on USLI (Ultra Large Scale means -more than about 1 million circuit elements on a single chip) Integration.
 - All the current computers were considered as 5th generation computers.
15. From the first generation to the fifth generation computers
- Processing power/speed/efficiency increased.
 - No of components/transistors per chip increased.
 - Storage capacity increased.
 - Power consumption/usage decreased.
 - Heat dissipation decreased.
 - Size decreased.
16. Unitwise MCQ questions can be done via https://ictmcq.home.blog/ol_ict_english/



Past paper questions

MCQ

- Which of the following was conceived by Charles Babbage that embodies the key features of modern computers. GIT2005
1)analytic engine 2)analytic computer 3)difference engine 4)tabulating machine
- Basic components of system GIT2005
1)Hardware, Software and Liveware
2)Strategic, Practical and Operational
3)Expert System, Decision Support and Transaction Support System
4)Input, Output and Processing
- Which of the following is considered as raw data? OL2007MCQ01
(1) Marks obtained by a student in a class.
(2) Sorted list of marks obtained by student in a class.
(3) Maximum marks obtained by students in a class
(4) Average marks obtained by student in a class
- First generation (IG/I Gen) computers were based on OL2007MCQ02
(1) Vacuum tubes (2) Transistor
(3) Integrated circuits (4) Microprocessors.
- An error in a computer program is referred to as a OL 2007
(1) Syntax. (2) Bug (3) debug (4) virus.
- Which of the following is **not** a value attribute of information? GIT 2007
1)accuracy 2)Size 3)Relevance 4)Timeliness
- Which of the following does **not** contain information? OL2008MCQ01
1)A weather report 2)A timetable
3)A price list displayed in a super market 4)An unlabeled chart
- Which of the following is **not** a basic component of a computer system? GIT 2008
1)Central Processing Unit 2)Input Unit 3)Output Unit 4)Networking Unit
- Which of the following statements is **not** true about eGovernment? GIT2008MCQ40
1)eGovernment deals with improving government services
2)eGovernment facilitates access to information on the government to citizens
3)eGovernment helps people get government services without need to travel long distance
4)eGovernmnet is a means for the government to generate funds to balance the national budget
- A doctor monitors the temperature of a patient having fever over a five hour period by: OL2009MCQ01
(A) Laying the hand on the patient's forehead at an hourly interval.
(B) asking the patient about the fever condition at an hourly interval
(C) drawing a graph using the readings obtained by a thermometer at an hourly interval.
Which of the above gives/gives the doctor, information about the patient's fever condition?
1)Only (A) 2)Only (B) 3)Only(C) 4)All (A), (B) and (C)

11. The invention of the Integrated Circuits marked the start of the Generation of computers OL2009MCQ02
1)first 2)second 3)third 4)fourth
12. Which of the following is **not** correct regarding the data and or information? GIT2009MCQ13
1)Data must be processed to produce Information
2)Complete, accurate, relevant and timely information can be obtained from processed data for effective decision making.
3)Information is produced by performing one or more operations on data.
4)Information must be processed to produce data.
13. "As vacuum tubes were replaced by transistors and transistors were subsequently replaced by microprocessors the A of the computers reduced."
Which of the following terms is appropriate for A ? OL2010MCQ01
1)capactiy 2)size 3)reliability 4)efficiency
14. "B is processed into C by information systems"
Which of the following pairs of icons is appropriate for B and C respectively? OL2010MCQ02
1)Information, Data 2)Data, Information
3)Data, Instructions 4)Instructions, information
15. Which of the following is **incorrect** information statement? GIT2010MCQ36
1)On-line shopping is possible using internet.
2)Doctors can use computers to analyse medical records of patients.
3)Computers and Internet can be used to obtain government services from home.
4)On-line shopping is possible without the use of communication link to the internet.
16. Which of the following can be classified as data in comparison with information? OL2011MCQ01
1)Average monthly temperature in the city of Anuradhapura.
2)A pie chart showing percentage of male and female students in a school.
3)Last four digits of a residential telephone number.
4)Maximum mark obtained for Mathematics by Grade 10 students.
17. Consider the following types of computer programming languages: OL2011MCQ26
A - Machine languages
B - Assembly languages
C - High level languages
Which of the following shows the order of generation of the above programming languages in ascending order?
1)A, B, C 2)B, C, A 3)C, B, A 4)C, A, B
18. The third generation of computers are based on the GIT2011MCQ05
1)Vacuum Tube 2)Transistors
3)Integrated Circuits 4)Very Large Scale Integrated (VLSI) circuit
19. Who invented Analytical Engine? AL2011MCQ01
1)Blaise Pascal 2)Charles Babbage 3)John Von Neumann
4)John V. Atanasoff 5)John Presper Eckert
20. Which of the following techniques is used in the Second Generation Computers? AL2011MCQ02
1)Integrated Circuits 2)Large Scale Integration 3)Microprocessors
4)Transistors 5)Vacuum Tubes
21. Computerized medical imaging systems which use echoes of transmitted pulses are called AL 2011
1)CAT scans 2)CT Scans 3)PETT scans 4)Ultra sounds scan 5) X-ray images
22. Which of the following represents the correct order of evolution in computer processor technology? OL2012MCQ01
1)Transistors, Integrated Circuits, Vacuum tubes
2)Transistors, Vacuum tubes, Integrated Circuits
3)Vacuum tubes, Integrated Circuits, Transistors
4)Vacuum tubes, Transistors, Integrated Circuits

23. In a library Information System, when a user enters the title of a book, it's availability is indicated after searching the database. Which of the following indicates the correct order for input, process and output of the system? OL2012MCQ02
- 1)book title, availability of the book, searching the database
 - 2)book title, searching the database, availability of the book
 - 3)searching the database, book title, availability of the book
 - 4)searching the database, availability of the book, book title
24. Consider the following: OL2012MCQ11
- A - Marks obtained by a student for mathematics
B - The average marks for mathematics of all students
- Which of the following is correct in relation to the above?
- 1)Both A and B represents Data
 - 2)Both A and B represents Information
 - 3)While A represents data, B represents information
 - 4)While A represents information, B represents data.
25. Which of the following types of information represents the correct order of evolution of the technology used in computer processors? GIT2012MCQ02
- 1)Transistors, Vacuum tubes, Integrated Circuit (IC), Very Large Scale Integration (VLSI)
 - 2)Vacuum tubes, Transistors, Integrated Circuit (IC), Very Large Scale Integration (VLSI)
 - 3)Vacuum Tube, Integrated Circuit (IC), Transistor, Very Large Scale Integration (VLSI)
 - 4)Vacuum Tube, Integrated Circuit (IC), Very Large Scale Integration (VLSI)< Transistor
26. Who is considered as the first computer programmer? AL 2012
- 1)John Von Neumann 2)Blaise Pascal 3)Charles Babbage 4)Ada Augusta Lovelace
 - 5)John Presper Eckert
27. Which of the following technologies has been used in the Third Generation Computers? AL 2012
- 1)Integrated Circuits (ICs) 2)Large Scale Integration (LSI) 3)Micro Processors
 - 4)Transistors 5)Vacuum Tubes
28. Consider the following statements about data and information. AL 2012
- A - The symbols '101011101'
B - Numbers, characters and images
C - Facts derived from a study
D - Facts that have been processed in such a way as to be meaningful to the person who receives it.
- Which of the above statements best describe/s "information"?
- 1)D only 2)A and B only 3)C and D only 4)A, B and C only 5)B, C and D only
29. Consider the following terms AL 2012
- A - input, B - output, C - Process, D - Storage
- Which of the above are **essential** for a system?
- 1)A and B only 2)A, B and C only 3)A, C and D only 4)B, C and D only 5)All A, B, C and D
30. Consider the following conversation of data into information OL 2013
- Data ->X->Information
- Which of the following is most suitable to replace X?
- 1)Displaying 2)Processing 3)Programming 4)Testing
31. With the development of technology the vacuum tubes were replaced by transistors. Which of the following is correct regarding the resultant changes to the computer? OL 2013
- 1)Speed decreased, power consumption increased and size decreased
 - 2)Speed decreased, power consumption increased and size increased
 - 3)Speed increased, power consumption decreased and size decreased
 - 4)Speed increased, power consumption decreased and size increased
32. Babbage's Difference Engine is based on AL 2013
- 1)mechanical technology 2)vacuum tube technology 3)transistor technology
 - 4)Integrated Circuit (IC) technology 5)Very Large Scale Integrated (VLSI) Circuit technology

33. In the evolution of computers, the vacuum tube was replaced by the transistor and the transistor was later replaced by the integrated circuit (IC). In this process Ⓐ of the computer increased while the Ⓑ decreased.

The suitable terms for Ⓐ and Ⓑ respectively. OL 2014

- 1)efficiency, physical size 2)efficiency, reliability 3)Physical size, efficiency
4)physical size cost

34. Which of the following statement(s) is/are true regarding Telemedicine? OL 2014
A - The internet or other network systems are used to maintain the connection between the doctor and the patient.
B - Patients in local medical institution with the network connections could be remotely examined by specialist doctors residing in distant locations.
C - Telemedicine could be used to improve the efficiency of clinical, medical, health education, administrative and research activities.
1)A only 2)A and B only 3)A and C only 4)All A, B and C

35. Which of the following can be considered as transaction that can be performed using the internet? OL 2014
A - Placing an order to send gift items to friends
B - Requests to reserve accommodation for a holiday
C - Placing an order for a preferred dress after a fit-on
1)A and B only 2)A and C only 3)B and C only 4)All A, B and C

36. Which of the following is/are true about the first generation computers? GIT 2014
A - Physical size of the computers was relatively large.
B - Vacuum tubes were used in circuitry
C - Machine Language was used for programming
1)A only 2)B only 3)A and C only 4)All A, B and C

37. One of the principal inventors of the Electronic Numerical Integrator and Computer (ENIAC) was - AL 2014
1)Blaise Pascal 2)Charles Babbage 3)John Von Neumann
4)Ada Augusta Lovelace 5)John Presper Eckert

38. Which of the following statements is correct with respect to the evolution of computing devices? AL 2014
1)Vacuum tubes were used by Blaise Pascal to build Pascaline.
2)The Pascaline is considered as a first generation computing device.
3)Computers built using vacuum tubes are considered as second generation computers.
4)Electronic Numerical Integrator and Computer (ENIAC) was built using vacuum tubes.
5)Apple I and II are two examples for second generation computers.

39. Which of the following statements is true with respect to programming languages? AL 2014
1)Machine languages belong to the second generation programming languages.
2)Assembly languages programs can run directly on any computer.
3)Assembly languages belong to the first generation programming languages.
4)Assembly languages is more human readable form of machine language.
5)Machine language programs can be translated into assembly languages by using assemblers.

40. Which of the following is/are useful to consider when purchasing a computer? OL 2015
A - brand reputation of the computer
B - vendor (seller) reputation
C - warranty period of the computer
1)A only 2)B only 3)A and B only 4)All A, B and C

41. Consider the following - GIT 2015
A - Hardware specification of the computer
B - Reputation of the seller.
C - Warranty period and quality of after sales service
Which of the above aspects are important in purchasing computer?
1) A and B only 2)A and C only 3)B and C only 4)All A, B and C

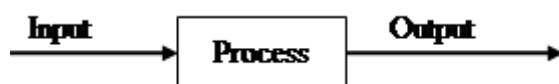
42. Charles Babbage is considered as the "father of the computer" by some people. This is because he AL 2015
- 1)invented the mechanical calculator Pascaline.
 - 2)invented the first re-programming electronic computing machine.
 - 3)took the leadership in building the first personal computer at IBM.
 - 4)introduced the concept of "Input, Process and Output" that is used in modern computers, for the first time.
 - 5)is the founder of the first electronic digital computer ENIAC (Electronic Numerical Integrator and Computer).
43. The first generation computers were based on AL 2015
- 1)Very Large Scale Integration (VLSI) technology.
 - 2)Large Scale Integration (LSI) technology.
 - 3)Integrated Circuits (Ics).
 - 4)Transistors
 - 5)Vacuum Tubes.
44. Consider the following statements OL 2016
- A - A student obtains medical advice from a doctor by contacting him over the telephone.
- B - An automated system, based on the humidity and temperature releases water to plants in a green house.
- C - A teacher purchases a book which is not available in Sri Lanka, from an online book shop.
- Which of the above use the Information and Communication Technology (ICT) effectively?
- 1)A and B only
 - 2)A and C only
 - 3)B and C only
 - 4)All A, B and C
45. What was the effect when vacuum tubes were replaced by transistors and later transistors were replaced by integrated circuits in the CPU? OL 2016
- 1)Both processing power and electricity consumption increased.
 - 2)Processing power increased while electricity consumption decreased.
 - 3)Processing power decreased while electricity consumption increased.
 - 4)Both processing power and electricity consumption decreased.
46. Nimal is writing a computer programme to create a graph to show the number of Dengue cases for the last five years. Input, process and output related to the programme are given in the steps A to D below but **not** in the correct order. OL 2016
- A - Show the graph for the total number of Dengue cases for the last five years
- B - Number of Dengue cases for each year
- C - Current year
- D - Calculate the total number of Dengue cases for the last five years
- Which of the following correctly shows the input, process and output?
- 1)Inputs: A and B process : C output:D
 - 2)Inputs: B and C process : D output:A
 - 3)Input: B process : C and D output:A
 - 4)Inputs: B and D process : A output:C
47. Consider the following statements: GIT 2016
- A - In a distance learning environment students can access learning materials from their home any time and attend to the lessons conveniently especially when they are unable to participate in traditional classroom teaching.
- B - Using modern technology such as mobile devices, the government can provide up to date information to farmers such as the possible crops to be cultivated, current prices of the crops etc.
- C - The modern technology facilitates the general public to reserve tickets in long distance trains and buses using mobile devices.
- Which of the above examples for e-services?
- 1)A and B only
 - 2)A and C only
 - 3)B and C only
 - 4)All A, B and C
48. Amarna finds out that, since 1960's, the cost of the computer hardware is considerably decreasing while the cost of software is comparatively increasing. She explains the reasons as: GIT 2016
- A - Advances in technology reduce the hardware cost.
- B - Increasing complexity of software increases the software cost.
- Which of the following is acceptable regarding the above reasons.
- 1)Both A and B are valid
 - 2)A is valid and B is invalid
 - 3)A is invalid and B is valid
 - 4)Both A and B are invalid

49. Which of the following statements is true? AL2016
- 1)The first generation of computers were built using transistors.
 - 2)Electronic Numerical Integrator and Computer (ENIAC) is a second generation computer.
 - 3)Ada Lovelace is the inventor of the analytical Engine.
 - 4)Alan Turing is considered as the first computer programmer.
 - 5)The first calculating device is believed to be the Abacus.
50. Consider the following scenarios: OL2017
- A - Obtaining a copy of an original birth certificate online
 - B - Paying water bills online using the website of the National Water Supply and Drainage Board.
 - C - Obtaining the revenue license for a vehicle services?
- 1)A and B only 2)A and C only 3)B and C only 4)All A, B and C
51. Which of the following are examples of e-government? GIT2017MCQ26
- A - A person applying for his vehicle revenue license online
 - B - Students accessing exam results from the Department of Examination website
 - C - A school library using an automated system to issue books to school children
- 1)A only 2)A and B only 3)B and C only 4)All A, B and C
52. Consider the following statements relating to online shopping. GIT2017MCQ29
- A - It saves travelling time spent for shopping.
 - B - Customers cannot physically examine a product before buying it.
 - C - Customers can find cheaper products easily.
- Which of the above are correct statements relating to online shopping?
- 1)A and B only 2)A and C only 3)B and C only 4)All A, B and C
53. A person inserts his Automatic Teller Machine (ATM) card into a bank ATM and enters the pin code. Then he enters the amount of money he wants to withdraw. After checking the available balance in the account, the ATM dispenses the money, updates the balance and displays the new balance. OL 2018.
- Which of the following, lists an input, a process, and output in the given scenario respectively?
- 1)amount of money to withdraw, new balance, money
 - 2)amount of money to withdraw, updating the balance, new balance
 - 3)new balance, pin code, amount of money to withdraw
 - 4)updating the balance, pin code, new balance
54. Which of the following occurred during the evolution of computer technology from the first generation to the current? OL 2018
- A -processing power of computers increased
 - B -number of transistors per square centimetre in the CPU increased
 - C -storage capacity of computers increased
- 1)A and B only 2)A and C only 3)B and C only 4)all A, B and C
55. Which of the following is an example for **not** an e-government service? OL 2018
- 1)Accessing GCE OL results from website of the Department of Examinations, Sri Lanka
 - 2)Buying items from an online shop
 - 3)Downloading National Identity Card application from the Department of Registration of Persons
 - 4)Paying water bills online on the website of the National Water Supply and Drainage Board
56. Consider the following statements related to development of computers over time. AL 2018
- A - Both processing speed and power consumption of computers have increased.
 - B - Processing speed of computer has increased while physical size of a computer has decreased.
 - C - Both power consumption and the physical size of a computer have reduced.
- Which of the above statements is/are valid?
- 1)A only 2)B only 3)A and B only 4)B and C only 5)All A, B and C
57. Which of the following devices have both input and output capabilities? OL 2019
- 1)joystick 2)OCR Device 3)touch screen 4)webcam

58. Which of the following is correct regarding the generations of computers? OL2019MCQ03
- 1) Integrated Circuits (ICs) were introduced in 1st generation computers.
 - 2) Vacuum tubes were introduced in 2nd generation computers.
 - 3) Transistors were introduced in 3rd generation computers.
 - 4) Very Large Scale Integration Circuits (VLSI) were introduced in 4th generation computers.
59. Which of the following are related to G2C (Government to Citizen) services in Sri Lanka? OL2019MCQ04
- A - viewing G.C.E (O/L) results online
 - B - ordering food items online.
 - C - renewing vehicle revenue licenses online.
- 1) A only 2) A and C only 3) B and C only 4) all A, B and C
60. Which of the following is an example for data processing? OL2019MCQ05
- 1) calculating the account balance using banking software after a cash withdrawal
 - 2) copying a file from a USB drive to a hard disk of a computer
 - 3) installing sound editing software in a personal computer
 - 4) scanning a letter using a scanner.
61. Which of the following made a significant contribution to the growth of Information and Communication Technology (ICT) usage? AL2019MCQ08
- A - exponential progress of the semiconductor technology paving the way for low cost hardware.
 - B - Introduction of user friendly software and interfaces to computers.
 - C - merge of computer and communication technologies to produce smart and mobile devices.
- 1) A only 2) B only 3) A and C only 4) B and C only 5) All A, B and C
62. Computer has evolved from the early main frames to the relatively small smart devices with high computing power used today. Which of the following inventions contributed to reduce the physical size of computers? AL2020MCQ03
- 1) bus 2) integrated circuits 3) registers 4) solid state memory 5) vacuum tube
63. Which of the following is an example for a data processing task of a user? OL2021MCQ01
- 1) Installing word processing software in a personal computer
 - 2) Formatting the hard disk in a personal computer
 - 3) Reading an online newspaper using a web browser
 - 4) Calculating the average marks obtained by a student using a spreadsheet software
64. Krishna accessed the official web portal of the government of Sri Lanka (<http://www.gov.lk>) to renew her vehicle revenue license online. Which of the following services was obtained by her? OL2021MCQ06
- 1) G2B 2) G2C 3) G2E 4) G2G

Paper 2

1. Describe the difference between data and information GIT 2006
2. The following figure shows the basic Input-Process-Output model used in Information Technology. Accordingly a system receives data its input and performs a meaningful process to deliver Information as the output. OL 2007



An information system has been developed to input income and expenses of a person under different categories of needs such as food, house rent, health etc. of a person. The system is to provide information on monthly total for each category of his expenses, the total expenditure of the month and saving of the month.

- (i) In addition to inputs given above, suggest another five inputs that can be entered into the system?
 - (ii) What calculations should be performed by the system in order to produce the above information?
 - (iii) Briefly explain, why the above system can be considered as an example of a batch processing system.
3. Write in one sentence, the difference between raw data and information. OL 2008
 4. 'Mobile doctor' is an ambulance Service, which provides emergency service for those who need immediate medical attention. Currently, they do not have use a computer based information system. Those who wish to receive services are required to register

with them.

Customers who need the service should call the 'Customer Service Manager'. After a short conversation he sends an ambulance with a doctor and nurse to the place where the patient is

After providing emergency treatment, the patient is taken to the nearest hospital.

(i) List separately, the data and information required by the Customer Services Manager in order

to provide the service.

(ii) List what additional data is required to prepare the invoice.

(iii) State what benefits they can obtain if they use a computer based information system?

5. List two developments in Information and Communication Technology which makes it easier for students to carry out their educational activities from home. OL 2009
6. Write two advantages of using Information and Communication Technology in processing and releasing G.C.E (Ordinary Level) Examination results.
7. Classify computers into two types according to their purposes. OL 2011
8. G.C.E(Ordinary Level) candidates can obtain their examination results from the website of the Department of Examinations. When the index number is entered to the data entry form of the website the system validates it. If it is valid, then the results will be retrieved from the database and displayed on the screen. Write the Input, Process and output for this scenario. OL 2011
9. (i)Telemedicine allow investigation, monitoring and management of patients from remote locations. - OL 2012
 - a. List one advantages of telemedicine in addition to the remote accessibility
 - b. List one challenge in using telemedicine.(ii) One of your friends offers a copy of an unlicensed software CD to you. Do you agree or disagree? Justify your answer.
(iii) Briefly explain two examples for computer controlled systems in agriculture.
(iv) "e-Learning helps one to learn at his/her own pace(speed) compared to traditional classroom-based learning"
Do you agree with this statement? Briefly explain your answer.
10. Computers can be classified into generations according to the main electronic component (technology) used in the Central Processing Unit (CPU). Copy the below table into your answer script and fill the second Column - GIT 2013

Generations	Main Electronic Components (Technology) used
First	
Second	
Third	

11. Lakmal travels around the country to sell computer parts and accessories. After each sale, he enters the sales details into his tablet computer using the touch screen. At the end of the day he views the total sales details. What is the (a) input device (b) output device used by lakmal?
12. (i) A learning (Education) Management System (LMS) is an application of ICT in education sector.
Write down two other application of ICT in the education sector - OL 2014
(ii) An LMS can be used to conduct the activities given in the following table more efficiently and conveniently.
Therefore, the three categories students, teachers and school administration are benefited.

Activity Number	Activity
1	Registration of students
2	Marking students attendance register
3	Issuing of certificates
4	Conducting preliminary tests to assess students' skills
5	Conducting school based assessments
6	Using multimedia enriched lessons for self studying
7	Using multimedia enriched lessons to classroom teaching
8	Ability to access preferred lesson at preferred time

Identify which categories is mostly benefited by using the LMS for each of the above activities and write the respective category against the activity number.

13. A tourist takes photographs of interesting places in Sri Lanka using the built-in camera of his tablet computer and saves the captured images on the hard disk of the device. Later, he edits the images using the touch screen and copies them to a USB drive. OL 2015
Identify and write down two input devices and two storage devices used in the above scenario.
14. A company uses its own file server to store electronic documents, spreadsheets and images. In order to reduce the cost it decides to store these files in cloud based (online) drive. OL 2016
(a) write down **one** benefit of using cloud based (online) drive in addition to reduce the cost.
(b) Write down **one** disadvantage of using the cloud based (online) drive.
15. Geetha uses an automated Teller Machine (ATM) to withdraw money from her bank account. First she inserts the ATM card into the machine and enters the secret pin code. Then she enters the amount of money she wants to withdraw. The system checks the balance available in her account and dispenses money as the balance available in her account is more than the amount of money Geetha wants to withdraw. At the same time the system updates the balance available by deducting the amount of money dispensed from the previous balance. From the above scenario, write down an example each for input, processing and output. OL 2016
16. Sama goes to a bookshop to buy 3 exercise books and 2 pens. After finding the items, she takes them to the counter where the sales assistant scans the bar code stickers attached to each item into the information system in the shop. Quantities of each item are also entered into the system. The system calculates the cost of each item and the total cost of all the items. Then the system prints the bill for the transaction.
Write down **one** example each for input, process and output of the information system.
Write down **one** example each for input, process and output of the information system. OL 2017 Compulsory.
17. The Colombo Weather Centre records daily rainfall values for one month.
Write down two examples for information that can be found by processing the above mentioned daily rainfall data. OL 2019 Compulsory.
18. It has been a common practice to use Learning Management System (LMS) to manage both schools and higher education institutes. Write down **two** facilities provided by an LMS for students.

Answers

1. analytic engine
2. Input, Output and Processing
3. Marks obtained by a student in a class.
4. Vacuum tubes
5. Bug
6. Size.
7. An unlabeled chart
8. Networking Unit
9. e-government is a means for the government to generate funds to balance the national budget
10. Only (C)
11. third
12. Information must be processed to produce data.
13. Size
14. Data, Information
15. On-line shopping is possible without the use of communication link to the internet.
16. Last four digits of a residential telephone number.
17. A, B, C
18. Integrated Circuits
19. Charles Babbage
20. Transistors
21. Ultra sounds scan
22. Vacuum tubes, Transistors, Integrated Circuits
23. book title, searching the database, availability of the book
24. While A represents data, B represents information
25. Vacuum tubes, Transistors, Integrated Circuit (IC), Very Large Scale Integration (VLSI)
26. Ada Augusta Lovelace
27. Integrated Circuits (ICs)
28. C and D only
29. A, B and C only
30. Processing
31. Speed increased, power consumption decreased and size decreased
32. mechanical technology
33. efficiency, physical size

34. All A, B and C
35. A and B only
36. All A, B and C
37. John Presper Eckert
38. Electronic Numerical Integrator and Computer (ENIAC) was built using vacuum tubes.
39. Assembly languages is more human readable form of machine language.
40. All A, B and C
41. All A, B and C
42. introduced the concept of "Input, Process and Output" that is used in modern computers, for the first time.
43. Vacuum Tubes
44. All A, B and C
45. Processing power increased while electricity consumption decreased.
46. Inputs: B and C process : D output:A
47. All A, B and C
48. Both A and B are valid
49. The first calculating device is believed to be the Abacus.
50. All A, B and C
51. A and B only
52. All A, B and C
53. amount of money to withdraw, new balance, money
54. all A, B and C
55. Buying items from an online shop
56. B and C only
57. touch screen
58. Very Large Scale Integration Circuits (VLSI) were introduced in 4th generation computers.
59. A and C only.
60. calculating the account balance using banking software after a cash withdrawal
61. All A, B and C
62. integrated circuits
63. Calculating the average marks obtained by a student using a spreadsheet software
64. G2C